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# Bounds, the weak law, and review

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Probability and Statistics for Engineers   Fall 2026

Monday, November 9

Markov, Chebyshev, and the finite-variance weak LLN

Wednesday, November 11

No lecture

Friday, November 13

Midterm 2 review

## Bounds from partial information

A hypothetical delay  $W$ , measured in minutes, satisfies

$$W \geq 0, \quad \mathbb{E}[W] = 6, \quad \text{Var}(W) = 9.$$

What can we say about  $P(W \geq 18)$  without a full probability law?

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A tail probability measures how often a threshold is exceeded.

## Markov's inequality

If  $Z \geq 0$  almost surely,  $\mathbb{E}[Z] < \infty$ , and  $a > 0$ , then

$$P(Z \geq a) \leq \frac{\mathbb{E}[Z]}{a}.$$

Every outcome in the tail contributes at least  $a$  to  $Z$ .

Derive the bound by comparing  $Z$  with  $a \mathbf{1}_{\{Z \geq a\}}$ .

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A useful probability bound is  $\min\{1, \mathbb{E}[Z]/a\}$ .

## Chebyshev's inequality

Suppose  $\mathbb{E}[X] = \mu$  and  $\text{Var}(X) = \sigma^2 < \infty$ . For  $a > 0$ ,

$$P(|X - \mu| \geq a) \leq \frac{\sigma^2}{a^2}.$$

Squared distance from the mean is a nonnegative random variable.

Apply Markov to  $Z = (X - \mu)^2$ . What is the new threshold?

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Use  $\min\{1, \sigma^2/a^2\}$  as an upper bound. Finite variance is essential here.

## One-sided and two-sided events

For the delay,  $\mathbb{E}[W] = 6$ ,  $\text{Var}(W) = 9$ , and  $W \geq 0$ .



$$\{W \geq 18\} \subseteq \{|W - 6| \geq 12\}.$$

Event inclusion justifies the one-sided bound. There is no factor of  $1/2$ .

## Mean and variance of an average

Let  $X_1, \dots, X_n$  be iid with mean  $\mu$  and finite variance  $\sigma^2$ .

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i.$$

Derive its mean and variance. Identify the step that uses independence.

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Linearity of expectation holds even when the observations are dependent.

## A finite-sample guarantee

lid measurements have mean  $\mu$  and variance at most 4 units<sup>2</sup>. How large should  $n$  be to guarantee

$$P(|\bar{X}_n - \mu| < 1/2) \geq 0.95?$$

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In general: if  $\sigma^2 \leq v$ ,  $0 < \delta < 1$ , and  $\varepsilon > 0$ , use  $n \geq \max\{1, \lceil v/(\delta\varepsilon^2) \rceil\}$ .

This certifies a sufficient sample size. The true distribution may need fewer observations.

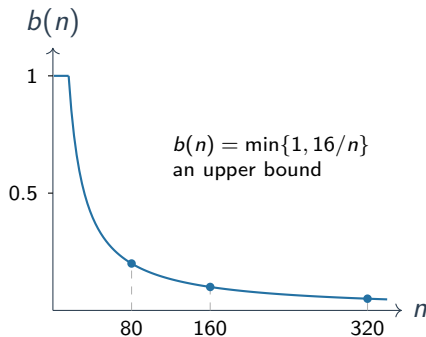
# Convergence in probability

For every fixed  $\varepsilon > 0$ ,

$$P(|Z_n - b| \geq \varepsilon) \longrightarrow 0.$$

We write  $Z_n \xrightarrow{P} b$ .

Fix the tolerated error first. Then let the sample size grow.



The curve connects upper bounds at positive integer sample sizes. It is not the actual tail probability.

## Weak LLN: statement and proof

Let  $X_1, X_2, \dots$  be iid with mean  $\mu$  and finite variance  $\sigma^2$ . Then

$$\overline{X}_n \xrightarrow{P} \mu.$$

Use the variance of the average in Chebyshev's inequality.

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This proof establishes the finite-variance weak law. It does not assume a normal distribution.

## Why dependence matters

Let  $P(Z = -1) = P(Z = 1) = 1/2$  and set  $X_i = Z$  for every  $i$ . Each observation has mean 0 and variance 1.

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n Z.$$

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Identical marginal distributions alone do not give the iid variance calculation.

## Hypotheses and interpretation

An iid average has variance at most  $4/n$ . At  $n = 320$ , Chebyshev gives

$$P(|\bar{X}_n - \mu| \geq 1/2) \leq 0.05.$$

State the corresponding success guarantee. Does the probability equal 0.95?

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The weak LLN concerns a probability limit, not a monotone path toward  $\mu$ .

## Midterm 2: scope and methods

Required material through November 9; proposed exam November 16. Working time: 55 minutes.

| Information       | Method and conclusion                                          |
|-------------------|----------------------------------------------------------------|
| A probability law | Sum or integrate over the event; obtain an exact value.        |
| Mean and variance | Chebyshev gives a bound when its hypotheses hold.              |
| iid observations  | The finite-variance weak LLN gives convergence in probability. |

Expectation, variance, joint/conditional laws, covariance, and elementary CDF transformations remain in scope.

CLT, inference, and stochastic processes are excluded.

## Expectation without independence

Twelve components each have failure probability 0.15. Let  $I_j$  indicate failure of component  $j$ , and let  $N = \sum_{j=1}^{12} I_j$ . Find  $\mathbb{E}[N]$ . Can we conclude that  $N$  is binomial?

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The method for a mean may need fewer assumptions than the full distribution.

## A joint probability table

| $p_{X,Y}(x,y)$ | $y = 0$ | $y = 1$ | $p_X(x)$ |
|----------------|---------|---------|----------|
| $x = 0$        | 3/8     | 1/8     |          |
| $x = 1$        | 1/8     | 3/8     |          |
| $p_Y(y)$       |         |         | 1        |

Find the marginals and  $P(Y = 1 \mid X = 1)$ . Are  $X$  and  $Y$  independent?

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Normalize the row selected by the conditioning event.

## Covariance in a linear combination

For the preceding table:

$$\mathbb{E}[X] = \mathbb{E}[Y] = \frac{1}{2}, \quad \text{Var}(X) = \text{Var}(Y) = \frac{1}{4}, \quad \mathbb{E}[XY] = \frac{3}{8}.$$

Find  $\text{Cov}(X, Y)$ ,  $\rho_{X,Y}$ , and  $\text{Var}(2X - Y)$ .

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A negative coefficient changes the sign of the covariance contribution.

## Exact exponential and normal probabilities

For  $T \sim \text{Exp}(2)$ , where 2 is a rate, find  $P(T > 2 \mid T > 1)$ .

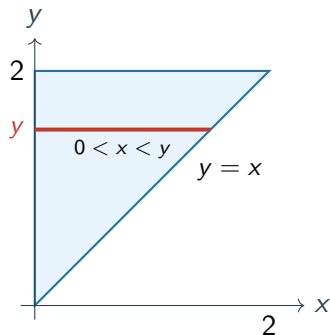
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For  $G \sim N(10, 4)$ , find  $P(8 < G < 12)$  in terms of  $\Phi$ .

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The stated normal model makes the second calculation exact.

## Probability over a triangular support



$$f_{X,Y}(x,y) = \begin{cases} 1/2, & 0 < x < y < 2, \\ 0, & \text{otherwise.} \end{cases}$$

Derive  $f_Y(y)$  and find  $P(Y < 1)$ .

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The triangle has area 2. Its density  $1/2$  therefore integrates to one.

## A conditional slice

In the same triangle,  $f_Y(y) = y/2$  for  $0 < y < 2$ . Find the conditional density of  $X$  given  $Y = y$ , then compute

$$P(X < 1 \mid Y = 3/2), \quad \mathbb{E}[X \mid Y = 3/2].$$

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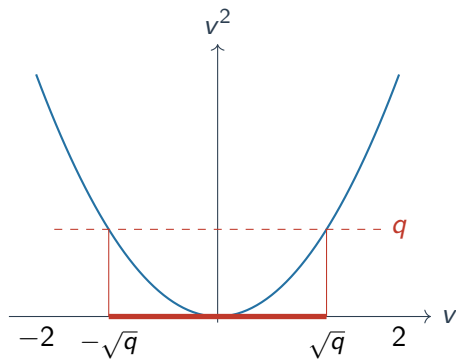
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The density ratio uses  $f_Y(y) > 0$ , not  $P(Y = y)$ .

## A transformation with two preimages



Let  $V \sim \text{Unif}(-2, 2)$  and  $Q = V^2$ . For  $0 \leq q < 4$ ,

$$\{Q \leq q\} = \{-\sqrt{q} \leq V \leq \sqrt{q}\}.$$

Find the full CDF and then the density.

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## A bound and a limit

lid measurements have mean  $\mu$  and variance at most 9.

- ① Bound  $P(|\bar{X}_{100} - \mu| \geq 1)$ .
- ② Find a sufficient  $n$  for failure probability at most 0.07.
- ③ Explain the limit for each fixed  $\varepsilon > 0$ .

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Round a sufficient integer sample size upward.

## Three statements to repair

- ❶ Zero covariance implies independence.
  - ❷ Differentiating a CDF always recovers its whole probability law.
  - ❸ The weak LLN makes every sample path move closer to  $\mu$  at every step.
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State exactly what the result establishes.

## Optional review: selecting a method

This block is ungraded and can be completed independently. Choose a method before calculating:

- ① A probability from a joint PMF, given an event in one coordinate.
  - ②  $P(8 < G < 12)$  when  $G \sim N(10, 4)$ .
  - ③ The law of  $g(X)$  when  $g$  is not one-to-one.
  - ④ A deviation probability when only mean and variance are supplied.
  - ⑤ The behavior of an iid average as  $n$  grows.
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## Optional review: table and covariance

| $p_{X,Y}(x,y)$ | $y = 0$ | $y = 1$ |
|----------------|---------|---------|
| $x = 0$        | $1/5$   | $3/10$  |
| $x = 1$        | $1/10$  | $2/5$   |

Find the marginals,  $P(X = 1 \mid Y = 1)$ ,  $\text{Cov}(X, Y)$ , and  $\text{Var}(X + Y)$ .

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## Optional review: a nonuniform joint density

Let  $f_{X,Y}(x,y) = \frac{2}{3}(x+2y)$  on  $0 < x < 1$ ,  $0 < y < 1$ , and zero elsewhere. Find  $f_Y$ ,  $f_{X|Y}$ , and  $P(X < 1/2 \mid Y = 1/2)$ .

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A rectangular support alone does not establish independence.

## Optional review: a decreasing transformation

Let  $U \sim \text{Unif}(0,1)$  and  $R = 1 - \sqrt{U}$ . Find its full CDF, its density, and  $P(R > 1/2)$ .

For  $0 \leq r \leq 1$ , translate  $R \leq r$  into an inequality in  $U$ .

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## Optional review: a sample-size certificate

lid observations have mean  $\mu$  and variance at most 5.

- ① At  $n = 80$ , bound  $P(|\bar{X}_n - \mu| \geq 1/2)$ .
- ② Find a sufficient integer  $n$  to make that probability at most 0.1.
- ③ Write the resulting success guarantee.

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## Optional review: the weak-LLN proof

State the finite-variance iid weak LLN and reconstruct its proof. Your argument must specify a positive tolerance and the moments of the average.

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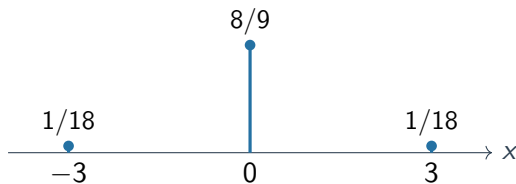
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## Optional: equality in Chebyshev's bound

Let  $X$  take the values  $-3, 0, 3$  with probabilities  $1/18, 8/9, 1/18$ .



Optional extension. Using a bound does not require proving that it is attained.

## Optional: exact tails and bounds

Let  $X \sim \text{Exp}(\lambda)$  and consider  $P(X \geq a/\lambda)$ ,  $a > 1$ .

| $a$ | Exact    | Markov | Chebyshev |
|-----|----------|--------|-----------|
| 2   | $e^{-2}$ | $1/2$  | 1         |
| 3   | $e^{-3}$ | $1/3$  | $1/4$     |

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Adapted from Harchol-Balter, supplied 2024 text, Exercise 18.14(a)–(c), p. 323.

# Reading selections

## **DeGroot and Schervish, fourth edition**

- p. 349: Markov and Chebyshev, Theorems 6.2.1–6.2.2 and proofs.
- p. 350: mean and variance of the average, Theorem 6.2.3.
- pp. 350–351: Example 6.2.1 only.
- p. 352: Definition 6.2.1 and Theorem 6.2.4; its proof on p. 353.

## **Harchol-Balter, supplied 2024 version**

- pp. 98–99: Theorems 5.16–5.17 and proofs.
- Optional: p. 323, Exercise 18.13 and Exercise 18.14(a)–(c).

These are selected passages. The slides and notes supply the required statements and proofs.